

AIM: To study network in Linux.

1.ifconfig command:

OUTPUT:

```
ubuntu@ubuntu:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd00::a00:27ff:fe0b:26c7 prefixlen 64 scopeid 0x0<global>
    inet6 fe80::a00:27ff:fe0b:26c7 prefixlen 64 scopeid 0x20<link>
    inet6 fd00::232a:c60:d214:881d prefixlen 64 scopeid 0x0<global>
    ether 08:00:27:0b:26:c7 txqueuelen 1000 (Ethernet)
    RX packets 3110 bytes 2195180 (2.1 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1974 bytes 233380 (233.3 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1529 bytes 123415 (123.4 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1529 bytes 123415 (123.4 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

2.ip addr:

ip -4 addr:

ip -6 addr:

OUTPUT:

```
ubuntu@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP
    group default qlen 1000
    link/ether 08:00:27:0b:26:c7 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 82958sec preferred_lft 82958sec
    inet6 fd00::232a:c60:d214:881d/64 scope global temporary dynamic
        valid_lft 86294sec preferred_lft 14294sec
    inet6 fd00::a00:27ff:fe0b:26c7/64 scope global dynamic mngtmpaddr
        valid_lft 86294sec preferred_lft 14294sec
    inet6 fe80::a00:27ff:fe0b:26c7/64 scope link
        valid_lft forever preferred_lft forever
```

```
ubuntu@ubuntu:~$ ip -4 addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 82894sec preferred_lft 82894sec
ubuntu@ubuntu:~$ ip -6 addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 state UNKNOWN qlen 1000
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 state UP qlen 1000
    inet6 fd00::232a:c60:d214:881d/64 scope global temporary dynamic
        valid_lft 86221sec preferred_lft 14221sec
    inet6 fd00::a00:27ff:fe0b:26c7/64 scope global dynamic mngtmpaddr
        valid_lft 86221sec preferred_lft 14221sec
    inet6 fe80::a00:27ff:fe0b:26c7/64 scope link
        valid_lft forever preferred_lft forever
```

3.netstat command:**OUTPUT:**

```
ubuntu@ubuntu:~$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
udp        0      0 10.0.2.15:bootpc       10.0.2.2:bootps        ESTABLISHED
udp        0      0 10.0.2.15:38060        10.0.2.3:domain        ESTABLISHED
udp        0      0 10.0.2.15:47028        10.0.2.3:domain        ESTABLISHED
udp        0      0 10.0.2.15:45955        10.0.2.3:domain        ESTABLISHED
udp        0      0 10.0.2.15:39834        10.0.2.3:domain        ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags               Type                   State         I-Node  Path
unix   2      [ ]                 DGRAM                9224
unix   3      [ ]                 STREAM              CONNECTED     11746    /run/dbus/system_bus_socket
unix   3      [ ]                 STREAM              CONNECTED     11509    /run/systemd/journal/stdout
unix   3      [ ]                 STREAM              CONNECTED     6524    /run/systemd/journal/stdout
unix   3      [ ]                 STREAM              CONNECTED     11243    /run/systemd/journal/stdout
unix   3      [ ]                 STREAM              CONNECTED     9715    /run/systemd/journal/stdout
```

netstat -a command:

```
ubuntu@ubuntu:~$ netstat -a
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 localhost:ipp           0.0.0.0:*               LISTEN
tcp        0      0 _localdnstproxy:domain 0.0.0.0:*               LISTEN
tcp        0      0 _localdnststub:domain  0.0.0.0:*               LISTEN
tcp6       0      0 ip6-localhost:ipp      [::]:*                 LISTEN
udp        0      0 _localdnstproxy:domain 0.0.0.0:*               ESTABLISHED
udp        0      0 _localdnststub:domain  0.0.0.0:*               ESTABLISHED
udp        0      0 10.0.2.15:bootpc       10.0.2.2:bootps        ESTABLISHED
udp        0      0 0.0.0.0:38018          0.0.0.0:*               ESTABLISHED
udp        0      0 0.0.0.0:mdns           0.0.0.0:*               ESTABLISHED
udp        0      0 0.0.0.0:631            0.0.0.0:*               ESTABLISHED
udp        0      0 localhost:59206         _localdnststub:domain  ESTABLISHED
udp        0      0 10.0.2.15:55203        10.0.2.3:domain        ESTABLISHED
udp        0      0 10.0.2.15:60360        10.0.2.3:domain        ESTABLISHED
udp6       0      0 [::]:34799             [::]:*                  ESTABLISHED
udp6       0      0 [::]:mdns              [::]:*                  ESTABLISHED
Active UNIX domain sockets (servers and established)
Proto RefCnt Flags               Type                   State         I-Node  Path
unix   2      [ ]                 DGRAM                9224
unix   3      [ ]                 STREAM              CONNECTED     11746    /run/dbus/system_bus_socket
unix   3      [ ]                 STREAM              CONNECTED     11509    /run/systemd/journal/stdout
unix   3      [ ]                 STREAM              CONNECTED     6524    /run/systemd/journal/stdout
unix   3      [ ]                 STREAM              CONNECTED     11243    /run/systemd/journal/stdout
unix   3      [ ]                 STREAM              CONNECTED     9715    /run/systemd/journal/stdout
unix   3      [ ]                 STREAM              CONNECTED     9105    /run/systemd/journal/stdout
unix   3      [ ]                 STREAM              CONNECTED     6761    /run/systemd/journal/stdout
unix   3      [ ]                 STREAM              CONNECTED     8070    /run/dbus/system_bus_socket
```

netstat -a T command:

```
ubuntu@ubuntu:~$ netstat -a T
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 localhost:ipp            0.0.0.0:*               LISTEN
tcp        0      0 _localdnstproxy:domain  0.0.0.0:*               LISTEN
tcp        0      0 _localdnststub:domain   0.0.0.0:*               LISTEN
tcp6       0      0 ip6-localhost:ipp       [::]:*                  LISTEN
udp        0      0 _localdnstproxy:domain  0.0.0.0:*
udp        0      0 _localdnststub:domain   0.0.0.0:*
udp        0      0 10.0.2.15:bootpc        10.0.2.2:bootps        ESTABLISHED
udp        0      0 0.0.0.0:38018           0.0.0.0:*
udp        0      0 0.0.0.0:mdns             0.0.0.0:*
udp        0      0 localhost:44661          _localdnststub:domain  ESTABLISHED
udp        0      0 0.0.0.0:631             0.0.0.0:*
udp        0      0 10.0.2.15:36544         10.0.2.3:domain        ESTABLISHED
udp        0      0 10.0.2.15:43875         10.0.2.3:domain        ESTABLISHED
udp        0      0 10.0.2.15:36544         10.0.2.3:domain        ESTABLISHED
udp        0      0 10.0.2.15:43875         10.0.2.3:domain        ESTABLISHED
udp6       0      0 [::]:34799              [::]:*
udp6       0      0 [::]:mdns                [::]:*

Active UNIX domain sockets (servers and established)
Proto RefCnt Flags       Type       State      I-Node  Path
unix   2      [ ]         DGRAM      9224
unix   3      [ ]         STREAM     CONNECTED  11746   /run/dbus/system_bus_socket
unix   3      [ ]         STREAM     CONNECTED  11509
unix   3      [ ]         STREAM     CONNECTED  6524    /run/systemd/journal/stdout
unix   3      [ ]         STREAM     CONNECTED  11243   /run/systemd/journal/stdout
unix   3      [ ]         STREAM     CONNECTED  9715    /run/systemd/journal/stdout
```

net -o command:

```
ubuntu@ubuntu:~$ netstat -o
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       Timer
tcp        0      0 10.0.2.15:41368         fracktail.canonica:http FIN_WAIT2   timewait (51.65/0/0)
udp        0      0 10.0.2.15:54311         10.0.2.3:domain        ESTABLISHED off (0.00/0/0)
udp        0      0 10.0.2.15:bootpc        10.0.2.2:bootps        ESTABLISHED off (0.00/0/0)
udp        0      0 10.0.2.15:36239         10.0.2.3:domain        ESTABLISHED off (0.00/0/0)
udp        0      0 10.0.2.15:60038         10.0.2.3:domain        ESTABLISHED off (0.00/0/0)
udp        0      0 10.0.2.15:44718         10.0.2.3:domain        ESTABLISHED off (0.00/0/0)
udp        0      0 10.0.2.15:53004         10.0.2.3:domain        ESTABLISHED off (0.00/0/0)

Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State      I-Node  Path
unix   2      [ ]         DGRAM      9224
unix   3      [ ]         STREAM     CONNECTED  11746   /run/dbus/system_bus_socket
unix   3      [ ]         STREAM     CONNECTED  11509
unix   3      [ ]         STREAM     CONNECTED  6524    /run/systemd/journal/stdout
unix   3      [ ]         STREAM     CONNECTED  11243   /run/systemd/journal/stdout
unix   3      [ ]         STREAM     CONNECTED  9715    /run/systemd/journal/stdout
unix   3      [ ]         STREAM     CONNECTED  9105
unix   3      [ ]         STREAM     CONNECTED  6761
unix   3      [ ]         STREAM     CONNECTED  8070    /run/dbus/system_bus_socket
```

net -u command:

```
ubuntu@ubuntu:~$ netstat -u
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
udp        0      0 10.0.2.15:bootpc        10.0.2.2:bootps        ESTABLISHED
udp        0      0 10.0.2.15:58641         10.0.2.3:domain        ESTABLISHED
udp        0      0 localhost:42699          _localdnststub:domain  ESTABLISHED
udp        0      0 10.0.2.15:53992         10.0.2.3:domain        ESTABLISHED
```

4.curl <url>:**OUTPUT:**

```
ubuntu@ubuntu:~$ curl https://amazon.in
<html>
<head><title>301 Moved Permanently</title></head>
<body>
<center><h1>301 Moved Permanently</h1></center>
<hr><center>Server</center>
</body>
</html>
```

5.curl -X POST <url>:**OUTPUT:**

```
ubuntu@ubuntu:~$ curl -X POST https://amazon.in
<html>
<head><title>301 Moved Permanently</title></head>
<body>
<center><h1>301 Moved Permanently</h1></center>
<hr><center>Server</center>
</body>
</html>
```

6.curl -X POST -data "p1=value1" <url>:**OUTPUT:**

```
ubuntu@ubuntu:~$ curl -X POST --data 'p1=ATOMIC' https://amazon.in
<html>
<head><title>301 Moved Permanently</title></head>
<body>
<center><h1>301 Moved Permanently</h1></center>
<hr><center>Server</center>
</body>
</html>
```

7.curl -X POST -d p1=value1 <url>:**OUTPUT:**

```
ubuntu@ubuntu:~$ curl -X POST -d "p1=ATOMIC" https://amazon.in
<html>
<head><title>301 Moved Permanently</title></head>
<body>
<center><h1>301 Moved Permanently</h1></center>
<hr><center>Server</center>
</body>
</html>
```

8.curl -o response <url>:**OUTPUT:**

```
ubuntu@ubuntu:~$ curl -o response https://amazon.in
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100  163      0  163    0     0    291      0 --:--:-- --:--:-- --:--:--   292
```

9.ping websitename.in/com:**OUTPUT:**

```
ubuntu@ubuntu:~$ ping amazon.in
PING amazon.in (52.95.116.115) 56(84) bytes of data.
64 bytes from 52.95.116.115: icmp_seq=1 ttl=255 time=170 ms
64 bytes from 52.95.116.115: icmp_seq=2 ttl=255 time=176 ms
64 bytes from 52.95.116.115: icmp_seq=3 ttl=255 time=170 ms
64 bytes from 52.95.116.115: icmp_seq=4 ttl=255 time=168 ms
64 bytes from 52.95.116.115: icmp_seq=5 ttl=255 time=168 ms
64 bytes from 52.95.116.115: icmp_seq=6 ttl=255 time=169 ms
64 bytes from 52.95.116.115: icmp_seq=7 ttl=255 time=169 ms
64 bytes from 52.95.116.115: icmp_seq=8 ttl=255 time=170 ms
64 bytes from 52.95.116.115: icmp_seq=9 ttl=255 time=170 ms
64 bytes from 52.95.116.115: icmp_seq=10 ttl=255 time=171 ms
64 bytes from 52.95.116.115: icmp_seq=11 ttl=255 time=168 ms
64 bytes from 52.95.116.115: icmp_seq=12 ttl=255 time=168 ms
64 bytes from 52.95.116.115: icmp_seq=13 ttl=255 time=168 ms
64 bytes from 52.95.116.115: icmp_seq=14 ttl=255 time=169 ms
64 bytes from 52.95.116.115: icmp_seq=15 ttl=255 time=193 ms
64 bytes from 52.95.116.115: icmp_seq=16 ttl=255 time=169 ms
64 bytes from 52.95.116.115: icmp_seq=17 ttl=255 time=169 ms
^C
--- amazon.in ping statistics ---
17 packets transmitted, 17 received, 0% packet loss, time 16017ms
rtt min/avg/max/mdev = 167.833/170.879/193.404/5.911 ms
```

10.hostname command:**OUTPUT:**

```
ubuntu@ubuntu:~$ hostname
ubuntu
```

11.traceroute websitename.in/com:**OUTPUT:**

```
ubuntu@ubuntu:~$ traceroute amazon.in
traceroute to amazon.in (52.95.116.115), 30 hops max, 60 byte packets
 1 _gateway (10.0.2.2) 12.357 ms 0.217 ms 0.493 ms
 2 * * *
 3 * * *
 4 * * *
 5 * * *
 6 * * *
 7 * * *
 8 * * *
 9 * * *
10 * * *
11 * * *
12 * * *
13 * * *
14 * * *
15 * * *
16 * * *
17 * * *
18 * * *
19 * * *
20 * * *
21 * * *
22 * * *
23 * * *
24 * * *
25 * * *
26 * * *
27 * * *
28 * * *
29 * * *
30 * * *
```

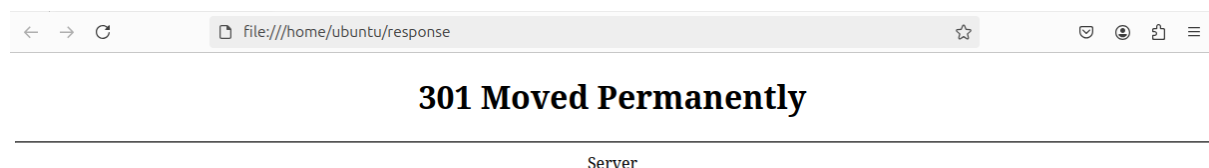
12.whois websitename.in/com:**OUTPUT:**

```
ubuntu@ubuntu:~$ whois amazon.in
Domain Name: amazon.in
Registry Domain ID: D15860-IN
Registrar WHOIS Server:
Registrar URL: http://www.markmonitor.com
Updated Date: 2025-01-15T09:50:52Z
Creation Date: 2005-02-11T11:14:14Z
Registry Expiry Date: 2026-02-11T11:14:14Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: registry.admin@markmonitor.com
Registrar Abuse Contact Phone: +1.2083895740
Domain Status: clientTransferProhibited http://www.icann.org/epp#clientTransferProhibited
Domain Status: clientDeleteProhibited http://www.icann.org/epp#clientDeleteProhibited
Domain Status: clientUpdateProhibited http://www.icann.org/epp#clientUpdateProhibited
Registry Registrant ID: REDACTED FOR PRIVACY
Registrant Name: REDACTED FOR PRIVACY
Registrant Organization: Amazon Technologies, Inc.
Registrant Street: REDACTED FOR PRIVACY
Registrant Street: REDACTED FOR PRIVACY
Registrant Street: REDACTED FOR PRIVACY
Registrant City: REDACTED FOR PRIVACY
Registrant State/Province: NV
Registrant Postal Code: REDACTED FOR PRIVACY
Registrant Country: US
Registrant Phone: REDACTED FOR PRIVACY
Registrant Phone Ext: REDACTED FOR PRIVACY
Registrant Fax: REDACTED FOR PRIVACY
```

```
Tech State/Province: REDACTED FOR PRIVACY
Tech Postal Code: REDACTED FOR PRIVACY
Tech Country: REDACTED FOR PRIVACY
Tech Phone: REDACTED FOR PRIVACY
Tech Phone Ext: REDACTED FOR PRIVACY
Tech Fax: REDACTED FOR PRIVACY
Tech Fax Ext: REDACTED FOR PRIVACY
Tech Email: Please contact the Registrar listed above
Name Server: ns2.amzndns.org
Name Server: ns2.amzndns.co.uk
Name Server: ns1.amzndns.net
Name Server: ns2.amzndns.net
Name Server: ns1.amzndns.co.uk
Name Server: ns1.amzndns.org
Name Server: ns1.amzndns.com
Name Server: ns2.amzndns.com
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of WHOIS database: 2025-03-08T03:32:33Z <<<

For more information on Whois status codes, please visit https://icann.org/epp
```

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8.Continuation:

Server